

Chih-Wei Tseng

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Education

National Yang Ming Chiao Tung University, Hsinchu, Taiwan

Feb. 2022 - Mar. 2024

M.S. in Electrical and Control Engineering

- GPA: 4.14/4.30
- Thesis: A Remote Photoplethysmography Atrial Fibrillation Detection System on Edge Devices
- Advisor: Chair Prof. Bing-Fei Wu
- Led a National Science and Technology Council (NSTC)-funded project on imaging-based AF and arrhythmia detection.
- Developed mobile-deployable, contact-free AF detection system achieving 99% reduction in model size, parameters, and FLOPs, 50% latency reduction, and >90% accuracy under motion and lighting variations.
- Applied GANs and diffusion models for motion-artifact denoising in rPPG signals.

Tamkang University, New Taipei City, Taiwan

Sep. 2018 - Jan. 2022

B.S. in Electrical and Computer Engineering

- Rank: 1/81; GPA: 3.97 / 4.00
- Thesis: A Poker Playing System Based on Deep Learning and Robotic Arms
- Advisor: Distinguished Prof. Ching-Chang Wong

Research Interest

Generative AI, Efficient AI, Interpretable AI, Robotics, Autonomous Vehicle, Health AI, RL

Research Experience

National Yang Ming Chiao Tung University, Hsinchu, Taiwan

Apr. 2024 – Mar. 2026

Research Assistant (Full-time)

- Developing generative models and classifiers to reconstruct ECG from PPG for multi-class arrhythmia detection.
- Leading three master's students on research on LLMs, generative AI, and autonomous driving.
- Building an interpretable mechanism of memory module in DL model for trajectory prediction, including rare-case synthesis (e.g. car accidents).

FaceHeart Inc., Taipei, Taiwan

Aug. 2025 – Dec. 2025

Research Intern (Part-time)

- Conducted research on deep learning interpretability for imaging-to-rPPG models.
- Explored visualization and analysis methods to improve model transparency and reliability.

Tamkang University, New Taipei City, Taiwan

July 2020 - Feb. 2022

Undergraduate Researcher

- Assisted senior-year student in building a data augmented system based on depth camera.
- Built YOLO-v4 based card recognition system achieving 98% accuracy.
- Developed a multi-agent reinforcement learning algorithm with a 70% game-winning rate.
- Integrated robotic arm control via a vision-based pose estimation interaction system.

Publication

C. -W. Tseng, B. -F. Wu* and Y. Sun, “A Real-Time Contact-Free Atrial Fibrillation Detection System for Mobile Devices,” in *IEEE Journal of Biomedical and Health Informatics*, vol. 29, no. 1, pp. 17-29, Jan. 2025, doi: 10.1109/JBHI.2024.3422155 (Feature Article).

C.-W. Sun, C.-W. Tseng, B.-F. Wu, “ CoSMIC: A Cognitively-Inspired System with Memory Interpretability and Consolidation in Autonomous Driving ” IEEE Transactions on Intelligent Transportation Systems, 2025. (Autonomous Grand Challenge 2025 – Innovation Award) (In Revision).

Teaching Experience

National Yang Ming Chiao Tung University, Hsinchu, Taiwan

July 2022 – June 2023

Teaching Assistant

- Delivered tutorials, graded assignments, and guided students in control-related labs and projects.

Awards

Autonomous Grand Challenge 2025 (ICCV Phase) – Innovation Award

Oct. 2025

- Received the Innovation Award among 19+ participating teams from both industry (including Nvidia, Bosch, Xiaomi) and academia.

IEEE Taipei Section Master’s Thesis Award

Dec. 2024

- Two master’s theses selected from ECE master’s theses from around Taiwan each year.

17th Topco Scientific Co., Ltd. (TSC) Thesis Award – Distinction

Nov. 2024

- 11 theses selected from 1,023 graduate students across all disciplines in Taiwan.

Presidential Award for Academic Excellence – Master’s Studies

Feb. 2023

Presidential Award for Academic Excellence – Undergraduate Studies * 5

Sep. 2018 – Jan. 2022

- Ranked in the top 1% of students in the class in a semester.

National Science and Technology Council Undergraduate Research Fellowship

July 2021 – June 2022

- Awarded to exceptional research proposals submitted by undergraduates across Taiwan; acceptance rate less than 30%.

3rd Place, Undergraduate Thesis Innovation Award

Dec. 2021

- Ranked 3rd place in undergraduate thesis competition among the department.

Outreach and Service

STEM Outreach Volunteer

July 2024

- Organized STEM workshops to promote hands-on robotics education in underserved communities.

Skills

Computer:

- Python, C/C++, Java, Kotlin, MATLAB, Verilog, LATEX, SolidWorks, Android Studio, ROS.

Languages:

- English: advanced (IELTS Overall 7.0, Listening 7.5, Reading 7.5, Writing 6.0, Speaking 6.0)
- Mandarin: native
- Japanese: beginner (N4)